SOC HOME LAB

Full Stack Edition · Wazuh · Shuffle · TheHive · Docker · VMware

PHASE 2

VM Deployment & Agent Setup Report

Prepared by **Anas TAGUI** · March 2026

1. Executive Summary

This report documents the execution of Phase 2 of the SOC Home Lab project: the deployment of three VMware-hosted virtual machines and the integration of all target endpoints into the Wazuh SIEM platform via agent installation. Phase 2 represents the transition from a purely containerised SOC backend to a fully realised lab environment in which attacker and victim machines operate on a dedicated virtual network and report security telemetry directly into the centralised SIEM.

Three virtual machines were deployed within VMware Workstation Pro: Kali Linux as the designated attacker machine, Windows Server 2022 as the primary Windows attack surface, and Metasploitable 3 as the intentionally vulnerable Linux target. A custom VMware NAT network was configured to provide static IP addressing and controlled inter-VM routing. Wazuh agents were installed and enrolled on both target machines, and Sysmon was deployed on Windows Server 2022 with the SwiftOnSecurity configuration to produce high-fidelity process and network telemetry.

> Milestone Achieved: Both target endpoints : Windows Server 2022 and Metasploitable 3 : are reporting as Active agents in the Wazuh dashboard. Sysmon events are visible in the SIEM, confirming rich telemetry collection is operational across the full lab environment.

2. Scope & Objectives

Phase 2 extends the lab environment from the Docker-only SOC platform into the virtual machine layer. The Docker stack deployed in Phase 1 continues to operate unchanged. This phase introduces the VMware virtualisation layer and establishes the network bridge between the two layers through which agent telemetry flows.

Primary Objectives

- Deploy and configure a dedicated VMware NAT network (192.168.100.0/24) with static IP assignments for all virtual machines.
- Import and configure the Kali Linux VMware OVA as the designated attacker endpoint.
- Build and configure the Windows Server 2022 virtual machine from the Microsoft evaluation ISO.
- Build and configure the Metasploitable 3 virtual machine as the intentionally vulnerable Linux target.
- Install and enrol the Wazuh agent on Windows Server 2022, pointing to the Wazuh Manager via the host bridge IP.
- Install and enrol the Wazuh agent on Metasploitable 3, confirming Linux agent connectivity.
- Deploy Sysmon on Windows Server 2022 using the SwiftOnSecurity configuration to enrich process and network event telemetry.
- Confirm both agents appear as Active in the Wazuh dashboard and that Sysmon events are visible in the SIEM event explorer.

3. Lab Network Architecture

The complete lab environment spans two isolated network layers that communicate through the Windows 11 host machine. The Docker layer hosts the SOC platform services; the VMware layer hosts the attacker and target endpoints.

Wazuh agents on the VMware VMs reach the Wazuh Manager container via the host machine's VMnet1 interface IP, which acts as the bridge between both layers.

Layer 1 — Docker Network (SOC Platform)

Service	Container Name	IP Address	Role
Wazuh Manager	wazuh-manager	172.20.0.10	Receives agent logs; processes detection rules; generates alerts
Wazuh Indexer	wazuh-indexer	172.20.0.11	OpenSearch — stores and indexes all security events
Wazuh Dashboard	wazuh-dashboard	172.20.0.12	Web UI — agent management, event exploration, alert review
Shuffle	shuffle	172.20.0.20	SOAR — automated alert-to-case workflow engine
TheHive	thehive	172.20.0.30	Incident case management platform

Layer 2 — VMware NAT Network (Endpoints)

Virtual Machine	IP Address	Operating System	Role
Kali Linux	192.168.100.30	Kali Linux (rolling)	Attacker : all offensive tooling (Nmap, Hydra, Metasploit, Mimikatz)
Windows Server 2022	192.168.100.20	Windows Server 2022 Evaluation	Target : Windows attack surface; Wazuh agent + Sysmon installed
Metasploitable 3	192.168.100.40	Ubuntu 14.04 (intentionally vulnerable)	Target : Linux attack surface; Wazuh agent installed

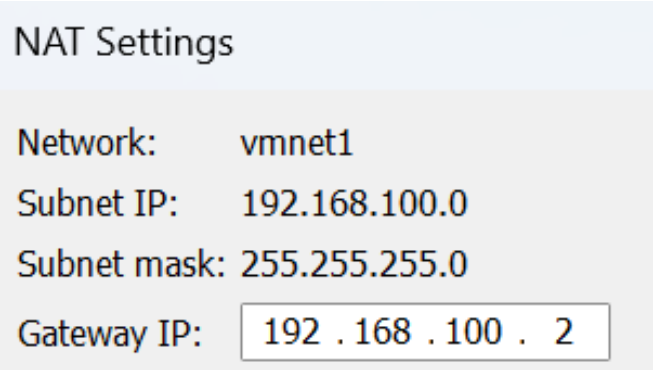
Bridge IP: The Wazuh Manager container is reachable from the VMware VMs at 192.168.100.1
the host machine's VMnet1 interface IP. Both Wazuh agents are configured with this address as their Manager endpoint, establishing the telemetry path between the two network layers.

4. Implementation

4.1 VMware Virtual Network Configuration

Prior to VM deployment, a custom NAT network was configured in VMware Workstation Pro using the Virtual Network Editor. A dedicated subnet (192.168.100.0/24) was assigned to VMnet1 to provide a controlled, isolated environment for all lab VMs. DHCP was disabled on this network in favour of static IP assignment at the OS level within each VM, ensuring addressing stability across reboots and eliminating dependency on VMware's DHCP service.

Parameter	Value
Network Adapter	VMnet1 (NAT)
Subnet	192.168.100.0/24
Gateway	192.168.100.2 (VMware NAT gateway)
Host Bridge IP	192.168.100.1 (VMnet1 interface : Wazuh Manager reachable here)
DHCP	Disabled : all VMs use static IP assignment
NAT External IP	Shared with Windows 11 host VMs have outbound internet access via NAT



4.2 Kali Linux VM Deployment

The Kali Linux VMware OVA, downloaded during Phase 0, was imported into VMware Workstation Pro via the Open a Virtual Machine workflow. No modifications to the OVA defaults were required at import time. Following import, the VM was powered on and the network adapter was reassigned to VMnet1.

A static IP address was configured within Kali's network settings to fix the machine at 192.168.100.30. Connectivity to the host and to the other VMs was confirmed via ICMP. Kali will serve exclusively as the attacker machine in Phase 3; no Wazuh agent is installed on this endpoint by design.

Parameter	Value
Allocated RAM	4 GB
vCPUs	2
Network Adapter	VMnet1 : Static IP 192.168.100.30
Wazuh Agent	Not installed — attacker machine
Primary Tools	Nmap, Hydra, Metasploit Framework, Mimikatz, Netcat

```
┌──(root㉿kali)-[/home/anas]
└─# sudo ip addr flush dev eth0

┌──(root㉿kali)-[/home/anas]
└─# sudo ip addr add 192.168.100.30/24 dev eth0

┌──(root㉿kali)-[/home/anas]
└─# sudo ip route add default via 192.168.100.2

┌──(root㉿kali)-[/home/anas]
└─# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default q
000
    link/ether 00:0c:29:dd:4a:dd brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.128/24 brd 192.168.100.255 scope global dynamic noprefixroute eth0
       valid_lft 1739sec preferred_lft 1739sec
    inet 192.168.100.30/24 scope global secondary eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fedd:4add/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(root㉿kali)-[/home/anas]
└─# ping 192.168.100.2
PING 192.168.100.2 (192.168.100.2) 56(84) bytes of data.
64 bytes from 192.168.100.2: icmp_seq=1 ttl=128 time=0.852 ms
```

4.3 Windows Server 2022 VM Deployment

Windows Server 2022 was provisioned from the Microsoft evaluation ISO downloaded during Phase 0. A new virtual machine was created in VMware Workstation Pro using the Typical configuration wizard, with the ISO attached as the installation media. The Standard Desktop Experience edition was selected during setup to provide a full GUI environment for agent administration and Group Policy configuration in Phase 4.

Following OS installation, VMware Tools was installed to enable clipboard sharing and display scaling. The network adapter was set to VMnet1, and a static IP of 192.168.100.20 was configured within Windows network settings. Windows Firewall was temporarily set to allow ICMP to confirm network connectivity across the lab.

Parameter	Value
Edition	Windows Server 2022 Standard : Desktop Experience
Allocated RAM	4 GB
vCPUs	2
Disk	60 GB dynamically allocated
Network Adapter	VMnet1 — Static IP 192.168.100.20
VMware Tools	Installed — clipboard and display scaling enabled

```
C:\Users\Administrator>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::28a2:7aeb:5343:de68%5
   IPv4 Address. . . . . . . . . . . : 192.168.100.20
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.100.2

C:\Users\Administrator>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4.4 Metasploitable 3 VM Deployment

Metasploitable 3 was built from the Rapid7 GitHub repository using the provided Vagrant and Packer build scripts. The Ubuntu 14.04-based build was selected as it provides the widest range of exploitable services for Phase 3 attack simulation. The build process produced a VMware-compatible VMDK image that was subsequently imported into VMware Workstation Pro.

The VM was assigned to VMnet1 and configured with a static IP of 192.168.100.40. The default Metasploitable credentials were documented and stored securely for Phase 3 reference. No hardening was applied to this machine; its intentionally vulnerable state is central to the attack simulation objectives of Phase 3.

Parameter	Value
Base OS	Ubuntu 14.04 LTS (intentionally unpatched)
Allocated RAM	2 GB
vCPUs	1
Network Adapter	VMnet1 : Static IP 192.168.100.40
Vulnerable Services	SSH, FTP (vsftpd backdoor), HTTP, SMB, PostgreSQL, and others
Hardening Applied	None : intentionally vulnerable by design

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:e6:32:d0
          inet addr:192.168.100.40  Bcast:192.168.100.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fee6:32d0/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:108 errors:0 dropped:0 overruns:0 frame:0
          TX packets:70 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:10103 (9.8 KB)  TX bytes:7225 (7.0 KB)
          Interrupt:17 Base address:0x2000
```

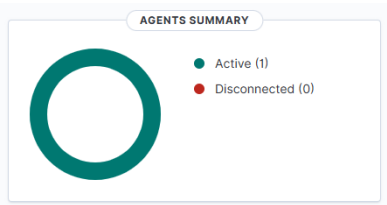
4.5 Wazuh Agent Installation : Windows Server 2022

The Wazuh agent MSI installer was downloaded directly from the Wazuh dashboard (Agents > Deploy New Agent). During installation, the Wazuh Manager address was set to 192.168.100.1 the host machine's VMnet1 bridge IP to route agent traffic from the VMware network into the Docker-hosted Manager container.

```
PS C:\Users\Administrator> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.0.msi -OutFi
wazuh-agent.msi
```

```
PS C:\Users\Administrator> msiexec /i wazuh-agent.msi WAZUH_MANAGER='192.168.100.1' /q
PS C:\Users\Administrator>
PS C:\Users\Administrator> NET START WazuhSvc
The requested service has already been started.
```

Agent enrolment was confirmed in the Wazuh dashboard within 30 seconds of service start. The agent status transitioned from Pending to Active. Initial event ingestion was visible in the Wazuh event explorer, confirming bidirectional communication between the Windows endpoint and the SIEM.



4.6 Sysmon Deployment : Windows Server 2022

Sysmon (System Monitor) was deployed on Windows Server 2022 to significantly enrich the quality of Windows telemetry forwarded to Wazuh. Sysmon captures process creation events, network connections, file creation timestamps, registry modifications, and driver loads events that are not available through standard Windows Event Log channels alone.

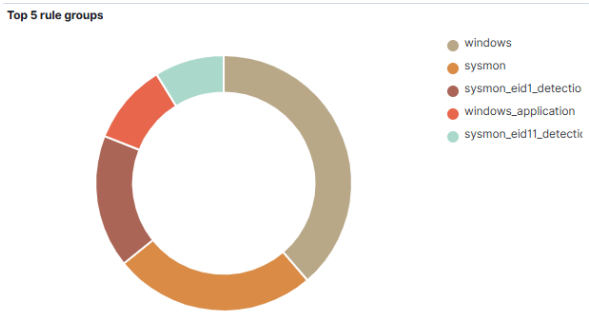
The SwiftOnSecurity Sysmon configuration was selected for its well-maintained ruleset that balances comprehensive coverage with noise reduction. The configuration file was downloaded and applied at Sysmon installation time.

```
PS C:\Users\Administrator> Invoke-WebRequest -Uri https://download.sysinternals.com/files/Sysmon.zip -OutFile C:\Users\A
dministrator\Desktop\Sysmon.zip
PS C:\Users\Administrator> Expand-Archive C:\Users\Administrator\Desktop\Sysmon.zip -DestinationPath C:\Users\Administra
tor\Desktop\Sysmon
PS C:\Users\Administrator> Invoke-WebRequest -Uri https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-config/master
/sysmonconfig-export.xml -OutFile C:\Users\Administrator\Desktop\Sysmon\sysmonconfig.xml
PS C:\Users\Administrator> & "C:\Users\Administrator\Desktop\Sysmon\Sysmon64.exe" -accepteula -i "C:\Users\Administrator
\Desktop\Sysmon\sysmonconfig.xml"


System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.90
Configuration file validated.
The service Sysmon64 is already registered. Uninstall Sysmon before reinstalling.
```

Verify it's running

```
PS C:\Users\Administrator> Get-Service Sysmon64

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon64           Sysmon64
```

Sysmon events (Windows Event Log channel: Microsoft-Windows-Sysmon/Operational) were confirmed visible in the Wazuh event explorer within minutes of installation. Event ID 1 (Process Creation), Event ID 3 (Network Connection), and Event ID 10 (Process Access) were all observed, confirming the configuration was active and forwarding correctly.



> Sysmon Event ID 10 : Process Access : is the event that will detect Mimikatz accessing the LSASS process in Phase 3. Its presence in the SIEM at this stage confirms the detection capability is in place ahead of the attack simulation phase.

4.7 Wazuh Agent Installation - Metasploitable 3

The Wazuh Linux agent was installed on Metasploitable 3 using the package manager installation method appropriate for the Ubuntu 14.04 base OS. The Manager IP was set to 192.168.100.1 to route telemetry through the same host bridge interface used by the Windows agent.

```
curl -so wazuh-agent.deb
https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.0_amd6
4.deb
WAZUH_MANAGER='192.168.100.1' dpkg -i ./wazuh-agent.deb
systemctl enable wazuh-agent && systemctl start wazuh-agent
```

Agent enrolment was confirmed in the Wazuh dashboard. The Metasploitable 3 agent transitioned to Active status and began forwarding syslog, authentication, and file integrity events into the SIEM. Both agents — Windows Server 2022 and Metasploitable 3 — were simultaneously visible as Active in the Wazuh Agents panel, confirming full dual-endpoint visibility.

4.8 Telemetry Baseline Review

Following agent installation and Sysmon deployment, a structured review of the incoming telemetry was conducted within the Wazuh event explorer. The purpose of this review was to establish a baseline understanding of normal activity across both endpoints before any attack simulation begins in Phase 3. Key observations included:

- Windows Server 2022: High volume of Sysmon process creation events (Event ID 1) corresponding to background Windows services. Authentication events (Windows Event ID 4624, 4625) visible for local and RDP login activity.
- Windows Server 2022: Sysmon network connection events (Event ID 3) capturing outbound connections from system processes. No anomalous external connections observed at baseline.
- Metasploitable 3: SSH authentication events forwarded via the Wazuh agent syslog decoder. FTP and HTTP service startup events visible. Existing Wazuh rules for Linux authentication failures confirmed active.

– Both agents: File integrity monitoring (FIM) baseline scans completed. Wazuh FIM module recorded initial file hashes for monitored directories on both endpoints.

Portfolio Note: The ability to articulate the difference between normal baseline telemetry and anomalous attack-phase events is a core SOC analyst competency. This baseline review ensures that Phase 3 detections can be discussed with reference to deviation from known-good behaviour.

5. Resource Allocation Summary

The following table documents the full memory allocation across the lab environment during Phase 2 operation. All three VMs and the Docker SOC stack are running simultaneously during the agent installation and validation steps.

Component	Allocated RAM	Runs On	Active During
Wazuh + Shuffle + TheHive (Docker)	4 GB	WSL2 on Windows host	All phases always on
Windows Server 2022 (VMware)	4 GB	VMware Workstation Pro	Phases 2–5 (lab sessions)
Kali Linux (VMware)	4 GB	VMware Workstation Pro	Phases 3–5 (attack sessions only)
Metasploitable 3 (VMware)	2 GB	VMware Workstation Pro	Phases 3–5 (attack sessions only)
Windows 11 Host OS	14 GB	Bare metal	Always
Headroom / Buffer	4 GB	Available	Available
Total (all running)	28 GB	32 GB installed	Phase 3 peak usage

6. Task Completion Matrix

The following matrix records the completion status of every deliverable scoped to Phase 2.

Task	Action Performed	Status
VMware NAT Network Configuration	VMnet1 configured subnet 192.168.100.0/24, DHCP disabled	**Completed**
Kali Linux OVA Import	OVA imported into VMware Workstation Pro	**Completed**
Kali Linux Network Configuration	Static IP 192.168.100.30 assigned on VMnet1	**Completed**
Kali Linux Connectivity Verification	ICMP confirmed to host and peer VMs	**Completed**

Windows Server 2022 VM Creation	Provisioned from evaluation ISO Standard Desktop Experience	**Completed**
VMware Tools Installation (Win Server)	Clipboard and display scaling enabled	**Completed**
Windows Server Network Configuration	Static IP 192.168.100.20 assigned on VMnet1	**Completed**
Metasploitable 3 Build	Built from Rapid7 Vagrant scripts VMware VMDK produced	**Completed**
Metasploitable 3 Network Configuration	Static IP 192.168.100.40 assigned on VMnet1	**Completed**
Wazuh Agent Install — Windows Server	MSI deployed Manager IP 192.168.100.1 configured	**Completed**
Wazuh Agent Active — Windows Server	Agent status: Active in Wazuh dashboard	**Completed**
Sysmon Installation — Windows Server	Sysmon64 deployed with SwiftOnSecurity config	**Completed**
Sysmon Events in Wazuh	Event IDs 1, 3, 10 confirmed visible in event explorer	**Completed**
Wazuh Agent Install — Metasploitable 3	DEB package deployed — Manager IP 192.168.100.1 configured	**Completed**
Wazuh Agent Active — Metasploitable 3	Agent status: Active in Wazuh dashboard	**Completed**
Dual-Agent Dashboard Screenshot	Both agents Active simultaneously committed to GitHub	**Completed**
Telemetry Baseline Review	Normal activity documented across both endpoints	**Completed**
Sysmon config committed to GitHub	sysmonconfig.xml saved to /phase-2-agents	**Completed**
Network diagram updated on GitHub	Architecture diagram updated to reflect VM layer	**Completed**

7. Issues Encountered & Resolutions

The following issues were identified during Phase 2 execution. Each was diagnosed and resolved before the phase was marked complete.

Issue	Root Cause	Resolution
Wazuh agent on Windows Server unable to connect to Manager	Windows Firewall was blocking outbound traffic on port 1514 (Wazuh agent communication port)	Added an outbound Windows Firewall rule permitting TCP/UDP traffic on port 1514 to destination 192.168.100.1

Metasploitable 3 build failing at Vagrant provisioning step	Vagrant version incompatibility with the Metasploitable 3 Vagrantfile on the build host	Downgraded Vagrant to the version specified in the Rapid7 build documentation; build completed successfully
Sysmon events not appearing in Wazuh event explorer	The Wazuh agent on Windows Server was not configured to collect from the Sysmon event log channel	Added the Microsoft-Windows-Sysmon/Operational log channel to the ossec.conf localfile block on the agent; restarted agent service
Metasploitable 3 agent showing Disconnected status intermittently	Metasploitable 3 OS clock drift causing TLS handshake failures with the Wazuh Manager	Installed and enabled ntpdate on Metasploitable 3 to synchronise system time; agent stabilised at Active status

8. Phase Outputs & Artefacts

The following deliverables were produced upon completion of Phase 2 and committed to the project repository.

- Screenshot: Wazuh dashboard showing both Windows Server 2022 and Metasploitable 3 as Active agents simultaneously — committed to /phase-2-agents/screenshots.
- Screenshot: Sysmon events (Event IDs 1, 3, 10) visible in the Wazuh event explorer — confirms high-fidelity telemetry collection operational.
- sysmonconfig.xml — SwiftOnSecurity Sysmon configuration file committed to /phase-2-agents.
- agents.md : documentation of each agent's configuration, monitored log channels, and collected event types, committed to /phase-2-agents.
- Updated network architecture diagram — reflects the full two-layer lab topology (Docker SOC platform + VMware VM layer), committed to /diagrams.
- This Phase 2 VM Deployment & Agent Setup Report — formal documentation of all deployment steps, network configuration, and validation outcomes.

9. Next Phase : Phase 3: Attack Simulation & Detection

With both target endpoints enrolled in the SIEM and high-fidelity Sysmon telemetry flowing, Phase 3 commences the attack simulation cycle. Five distinct attack techniques will be executed from Kali Linux against the target VMs. For each attack, the full SOC pipeline detection in Wazuh, automation in Shuffle, and case creation in TheHive will be exercised. Custom Wazuh detection rules will be authored for any technique not covered by the default ruleset.

Prepared by: **Anas TAGUI** · SOC Home Lab Project · March 2025

This document is part of a personal cybersecurity portfolio project. All tools used are open-source and free. The environment is self-hosted on personal hardware for educational and professional development purposes.